



Modelling Financial Markets

Chapter 8: Backtesting and Evaluating Risk Forecasts



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Acronyms used in this material (1/2)

- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BCBS: Basel Committee on Banking Supervision
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DGP: Data-Generating Process
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ EGARCH: Exponential GARCH
- ▣ ES: Expected Shortfall
- ▣ ESR: Expected Shortfall Regression (backtest)
- ▣ EVT: Extreme Value Theory
- ▣ FDR: False Discovery Rate
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FWER: Family-Wise Error Rate
- ▣ FZ: Fissler–Ziegel (family of scoring functions for VaR and ES)
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GPD: Generalised Pareto Distribution

Acronyms used in this material (2/2)

- ▣ HS: Historical Simulation
- ▣ IMA: Internal Models Approach
- ▣ LLM: Large Language Model
- ▣ MCS: Model Confidence Set
- ▣ OLS: Ordinary Least Squares
- ▣ PIT: Probability Integral Transform
- ▣ POF: Proportion Of Failures (test)
- ▣ QLIKE: Quasi-LIKElihood (loss function)

Chapter Route and Outcomes

Three groups of questions

- ▣ **Calibration**
 - ▶ is a VaR or ES forecast right, day after day?
 - ▶ can a backtest tell?
- ▣ **Comparison**
 - ▶ which of several forecasts is better?
 - ▶ is the gap significant?
- ▣ **Decision:** what can we conclude when the tail holds only a few days a year?
- ▣ Two parts
 - ▶ Part I: setup, VaR tests and their power, regulation, first ES tests
 - ▶ Part II: ES evidence, scores, comparative backtests, conformal VaR, case studies
- ▣ Technical detail sits in the [Appendix](#) at the end

After this chapter you can

- ▣ test the number and the timing of VaR breaches and state the size and power of each test at your T
- ▣ backtest ES with residuals, Z_2 by simulation and PIT-based statistics
- ▣ rank (VaR, ES) forecasts with a consistent score, DM tests and the MCS
- ▣ judge what a distribution-free (conformal) VaR does and does not guarantee

Setup and Notation (1/2)

- **Backtesting**: compare risk forecasts made with the information $\mathcal{F}_{t-1}$ with the outcomes that followed; $\mathcal{F}_{t-1}$: everything known at the close of day $t - 1$
- Daily log return $X_t = r_t = \ln(P_t/P_{t-1})$, P_t : closing price on day t ; daily **loss** $L_t = -r_t$ (positive number = money lost)
- **VaR** (Value-at-Risk): the loss exceeded with probability α , minus the α -quantile q_α of the return ([Chapter 7](#))
 - ▶ $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$; inf: the smallest x at which the distribution function exceeds α
- **ES** (Expected Shortfall): the average of the quantiles q_u over all tail levels $u \leq \alpha$, i.e. the mean loss beyond VaR
 - ▶ $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = E[L_t \mid L_t > \text{VaR}_\alpha]$ (continuous case)
 - ▶ Levels are tail probabilities: VaR 1% ($\alpha = 0.01$), VaR 2.5% and ES 2.5% ($\alpha = 0.025$); Basel used VaR 1% over 10 days, FRTB uses ES 2.5%, backtests use one-day forecasts

Setup and Notation (2/2)

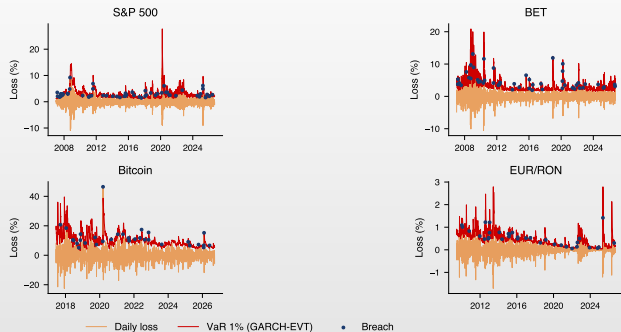
- A **breach** (exception) occurs when the loss exceeds the forecast
 - ▶ $I_t = 1\{L_t > \text{VaR}_t\}$, with target rate α ; $1\{\cdot\}$: indicator, 1 if the condition holds, 0 otherwise; VaR_t : the forecast made on day $t - 1$ for day t
- Ideal **conditional calibration**: whatever we knew yesterday, today's breach probability is exactly α (Christoffersen (1998))
 - ▶ $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$ for all t , so the I_t are i.i.d. Bernoulli(α): independent coin flips that equal 1 with probability α
- Two testable consequences
 - ▶ **Unconditional coverage**: about αT breaches in T days
 - ▶ **Independence**: breaches do not cluster

Six Competing Forecasters and the Experiment

Model	One-day VaR and ES from
HS	the empirical quantile of order $1 - \alpha$ of the last 1000 losses
Normal	the Normal distribution with the window mean and standard deviation
Student-t	a Student-t fitted to the window (location, scale, degrees of freedom ν)
GARCH-t	$\sigma_t^2 = \omega + a\varepsilon_{t-1}^2 + b\sigma_{t-1}^2$: variance = constant + reaction a to yesterday's squared shock ε_{t-1} + persistence b ; Student-t shocks
FHS	GARCH volatility $\times$ empirical quantiles of $z_t = \varepsilon_t/\sigma_t$
GARCH-EVT	GARCH volatility $\times$ a GPD (generalised Pareto) tail fitted to the largest 10% of standardised losses (McNeil & Frey (2000))

- Four series, each on its own calendar
 - ▶ S&P 500 and BET (Bucharest Stock Exchange index): closes
 - ▶ Bitcoin (trades every day); EUR/RON: BNR reference rate
- Rolling window of 1000 observations
 - ▶ each forecast uses data up to the previous day only
- Refits every 20 days
 - ▶ GARCH, Student-t and EVT parameters; volatility updated daily
- Evaluation days T , until 18.09.2026
 - ▶ S&P 500 4 959 and BET 4 937 (from 2007), EUR/RON 4 352 (from 2009), Bitcoin 3 384 (from 2017)

Four Markets, One Risk Model

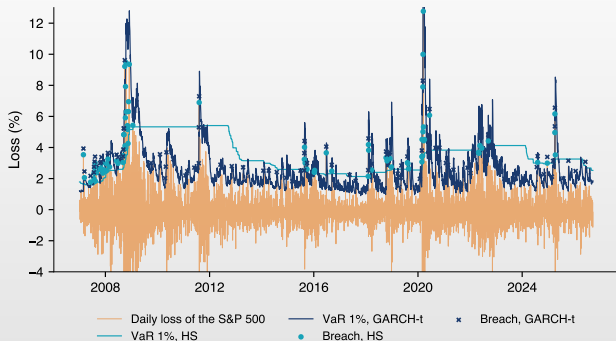


- Orange: daily losses; red: VaR 1% forecast by GARCH-EVT; blue dots: breaches ($L_t > VaR_t$)
- The VaR moves with volatility: it doubles or triples in 2008, March 2020 and the 2022 crypto crash
- Breach rates: S&P 500 1.13%, BET 0.99%, Bitcoin 0.95%, EUR/RON 0.78%
- EUR/RON has a much smaller scale: a managed exchange rate, long calm spells, rare jumps
- Question for the chapter: is a rate close to 1% enough to call a model good?

What a Backtest Can and Cannot Identify

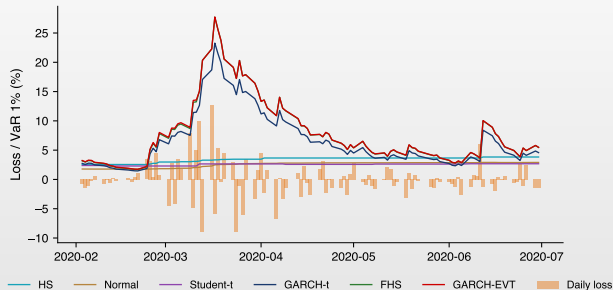
- Each backtest checks one implication of conditional calibration
 - ▶ Kupiec: the breach rate; Christoffersen: first-order memory
 - ▶ duration tests: the spacing of breaches; DQ: predictability from chosen regressors
- H_0 is a **joint** hypothesis on the whole forecasting procedure
 - ▶ distribution, volatility dynamics, estimated parameters, window length and refit rule
 - ▶ A rejection does not say which component failed
- One outcome per day: evidence accumulates only over many days
 - ▶ the textbook χ^2 limits treat VaR_t as known
 - ▶ estimation risk: last slide of the section “Backtesting VaR: Coverage and Independence”
- Non-rejection is not validation: it is informative only if the test has power against the relevant alternatives at the available T

S&P 500: Two VaR Forecasts, 2007–2026



- VaR 1%: HS in teal, GARCH-t in blue; orange: daily losses; markers: breaches of each model
- HS is a step function
 - ▶ it jumps only when extreme days enter or leave the 1000-day window
 - ▶ GARCH-t rises the day after volatility rises
- Almost the same count: HS 83, GARCH-t 88 breaches in 4959 days
- Similar counts can hide different timing
 - ▶ HS breaches arrive in bursts (2008, 2020)
 - ▶ so we must also test **when** breaches occur

March 2020: The Stress Test Nobody Planned

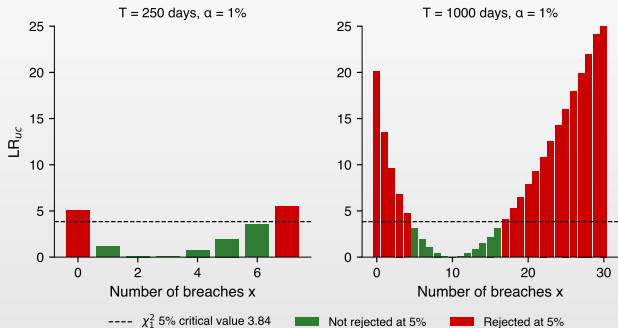


- S&P 500, February–June 2020: daily losses (bars) and the VaR 1% of the six models
- Largest loss: 12.77% on 16.03.2020
 - ▶ a correct VaR 1% expects 1.04 breaches in these 104 days
- Breaches observed
 - ▶ unconditional: HS 12, Normal 17, Student-t 17
 - ▶ conditional: GARCH-t 3, FHS 2, GARCH-EVT 2
- Current volatility mattered more than the tail shape
 - ▶ a fat tail without a moving scale (Student-t) did not help

Kupiec (1995): the POF Test

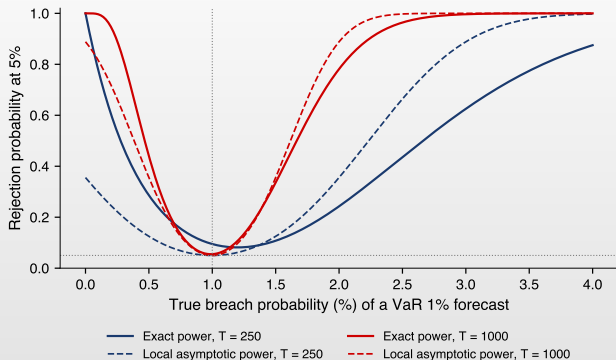
- Data: $x = \sum_t I_t$ breaches in T days; estimated breach rate $\hat{\pi} = x/T$
- H_0 : $\pi = \alpha$ (correct unconditional coverage), against H_1 : $\pi \neq \alpha$; π : the true breach probability
- Likelihood ratio statistic (Kupiec (1995)): compares the Binomial likelihood of the observed count under α with that under the observed rate $\hat{\pi}$
 - ▶ $LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x} \alpha^x}{(1-\hat{\pi})^{T-x} \hat{\pi}^x} \xrightarrow{d} \chi_1^2$
 - ▶ $LR_{uc} \geq 0$, equal to 0 when $\hat{\pi} = \alpha$; it grows as $\hat{\pi}$ moves away from α ; $\xrightarrow{d} \chi_1^2$: for large T its distribution under H_0 is chi-square with 1 degree of freedom
- Reject at 5% if $LR_{uc} > 3.84$; two-sided: too few breaches also count as failure; for small T use the exact Binomial(T, α) distribution of x
- Worked example, S&P 500, GARCH-t, 2008: $x = 8$ in $T = 253$ (expected 2.53): $LR_{uc} = 7.60$, $p = 0.0058$; exact Binomial $p = 2P(X \geq 8) = 0.009$

How Many Breaches Are Acceptable?



- LR_{uc} against the number of breaches x , one year (left) and four years (right) of VaR 1%
- $T = 250$: the test accepts x from 1 to 6; $T = 1000$: from 5 to 16
- Even $x = 0$ is rejected in one year
 - ▶ $LR_{uc} = -500 \ln 0.99 = 5.03$: a VaR never breached is too conservative
- One year holds about 2.5 informative days for VaR 1%
 - ▶ use long samples, two levels ($\alpha = 1\%$ and 2.5%) and several tests

Power of the POF Test

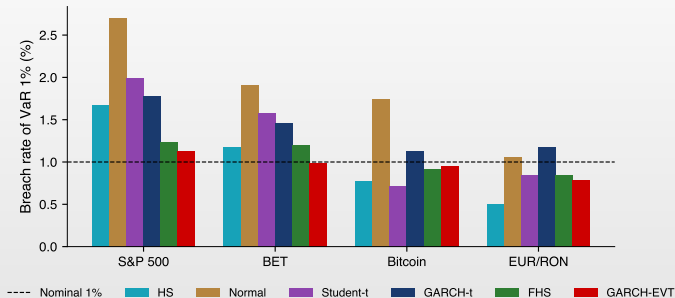


- Rejection probability at 5% against the true breach rate
 - ▶ solid: exact Binomial; dashed: local $\chi^2_1(\lambda)$ approximation (next slide)
- $T = 250$, true rate $\pi = 2\%$
 - ▶ local power 36%, exact power 24%
 - ▶ a doubled breach rate passes three years in four
- The rule $LR_{uc} > 3.84$ has actual size 9.5% at $T = 250$
 - ▶ $P(x = 0) = 8.1\%$ alone; not an exact 5% test
- 50% power needs
 - ▶ $\pi \approx 2.7\%$ at $T = 250$, $\pi \approx 1.7\%$ at $T = 1000$

Power of the POF Test: Local Alternatives

- Local alternatives: the true rate approaches α as T grows
 - ▶ $\pi_T = \alpha + h/\sqrt{T}$, h : a fixed distance on the $\sqrt{T}$ scale
- Under these alternatives the POF statistic is approximately non-central χ^2
 - ▶ $LR_{uc} = \frac{T(\hat{\pi} - \alpha)^2}{\alpha(1 - \alpha)} + o_p(1) \xrightarrow{d} \chi_1^2(\lambda), \quad \lambda = \frac{T(\pi - \alpha)^2}{\alpha(1 - \alpha)}$
 - ▶ $o_p(1)$: a term that vanishes in probability as $T \rightarrow \infty$
 - ▶ λ : non-centrality; it grows with T and with the squared miss $(\pi - \alpha)^2$
- Reading
 - ▶ power = $P(\chi_1^2(\lambda) > 3.84)$: larger λ , more power
 - ▶ $T = 250$, $\pi = 2\%$: $\lambda = 2.53$; one year of a doubled rate gives a small λ

Breach Rates Across Models and Markets

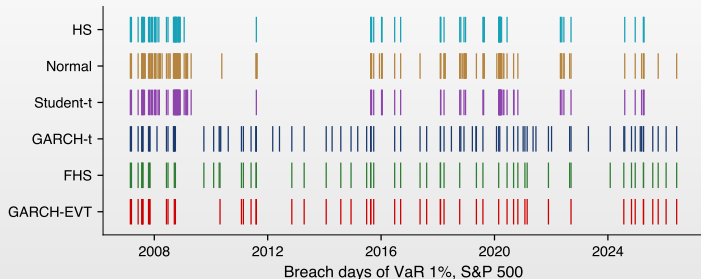


- Share of days with $L_t > \text{VaR}_t$ for VaR 1%; the Normal distribution fails except on EUR/RON: S&P 500 2.70%, BET 1.90%, Bitcoin 1.74%
- Same GARCH-t volatility, different tail: GARCH-t 1.77% and 1.46% versus FHS 1.23% and 1.20% (S&P 500, BET): evidence against the symmetric t tail on the loss side
- HS is too conservative on EUR/RON (0.51%): an old crisis in the window keeps its VaR high

Christoffersen (1998): Testing Independence

- ▣ Model I_t as a first-order Markov chain with $\pi_{ij} = P(I_t = j \mid I_{t-1} = i)$
- ▣ Count transitions n_{ij} (from state i yesterday to state j today); $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$
- ▣ H_0 : $\pi_{01} = \pi_{11} = \pi$ (no memory), with $\hat{\pi} = (n_{01} + n_{11}) / \sum n_{ij}$
- ▣ Statistic: numerator = likelihood with one breach probability; denominator = likelihood with a probability that depends on yesterday's state
 - ▶ $LR_{ind} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi_1^2$
 - ▶ Large when $\hat{\pi}_{11}$ (a breach after a breach) differs from $\hat{\pi}_{01}$ (a breach after a calm day); reject at 5% above 3.84
- ▣ Conditional coverage: $LR_{cc} = LR_{uc} + LR_{ind} \xrightarrow{d} \chi_2^2$ tests both properties at once (2 degrees of freedom, 5% critical value 5.99)

When Do Breaches Happen?

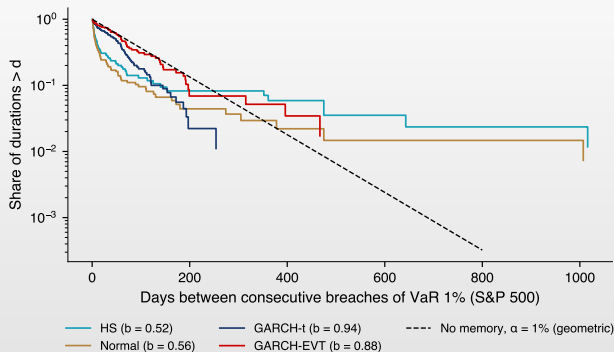


- Each tick: one breach of VaR 1% on the S&P 500; HS, Normal, Student-t: bursts in 2008, 2015–2016, 2018, 2020, 2022; filtered models: spread almost uniformly
- Counterexample, HS in 2007–2010 only: $n_{00} = 928$, $n_{01} = 38$, $n_{10} = 38$, $n_{11} = 3$; $\hat{\pi}_{11} = 0.0732$ versus $\hat{\pi}_{01} = 0.0393$, yet $LR_{ind} = 0.95$, $p = 0.33$: $\hat{\pi}_{11}$ rests on 3 pairs
- Full sample: independence rejected for HS ($p < 0.001$), Normal ($p < 0.001$) and, narrowly, for FHS ($p = 0.046$), not for GARCH-t ($p = 0.095$): a subsample non-rejection is not a full-sample verdict

Duration-Based Tests

- Duration D_i : number of days between breach $i - 1$ and breach i
- Under a correct VaR, breaches have no memory: D_i is geometric with mean $1/\alpha$ (100 days for $\alpha = 1\%$)
- Christoffersen & Pelletier (2004): approximate the discrete durations by a continuous Weibull model
 - ▶ $f(D; a, b) = a^b b D^{b-1} e^{-(aD)^b}$: the Weibull density of a duration, $a > 0$: rate (inverse scale), $b > 0$: shape; $b = 1$ is the exponential (memoryless) case
 - ▶ $b < 1$: short durations are too frequent, i.e. breaches cluster
- H_0 : $b = 1$ with a free, a likelihood ratio test (χ^2_1) of memorylessness, not of the mean $1/\alpha$ (coverage is left to Kupiec); first and last durations censored
- Uses the whole spacing pattern, not only consecutive days: more power against clustering

How Long Between Breaches?

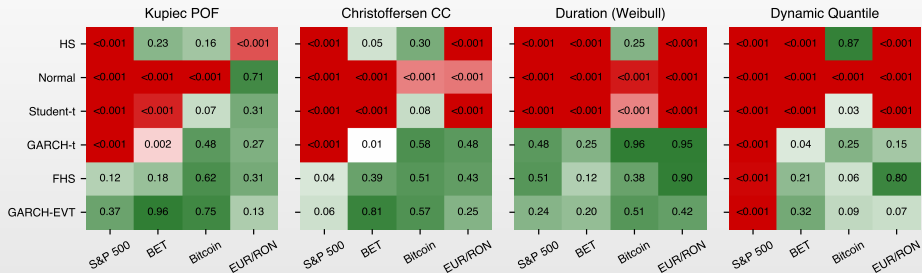


- Share of durations longer than d (log scale); dashed: no memory with $\alpha = 1\%$
- HS: $b = 0.52$, median 7 days, 44% of durations at most 5 days; test $p < 0.001$
- Normal: $b = 0.56$
 - ▶ GARCH-t: $b = 0.94$, median 48 days, $p = 0.479$
 - ▶ GARCH-EVT: $b = 0.88$, $p = 0.240$
- The long tail of HS durations: years without breaches after a crisis leaves a high VaR in the window

Regression-Based Backtests: the DQ Test

- Under correct conditional coverage $\text{Hit}_t = I_t - \alpha$ (the breach indicator, centred) is a martingale difference, i.e. unpredictable from the past: $E[\text{Hit}_t | \mathcal{F}_{t-1}] = 0$, $\text{Var}(\text{Hit}_t | \mathcal{F}_{t-1}) = \alpha(1 - \alpha)$
- So no $\mathcal{F}_{t-1}$ -measurable variable may predict it: regress $\text{Hit}_t = X_t\beta + u_t$ with $X_t = (1, \text{Hit}_{t-1}, \dots, \text{Hit}_{t-4}, \text{VaR}_t)$ and test $\beta = 0$
- Dynamic Quantile statistic ([Engle & Manganelli \(2004\)](#), Wald form with the null variance)
 - ▶ $\text{DQ} = \frac{\hat{\beta}' X' X \hat{\beta}}{\alpha(1 - \alpha)} = \frac{\text{Hit}' X (X' X)^{-1} X' \text{Hit}}{\alpha(1 - \alpha)} \xrightarrow{d} \chi_6^2$
 - ▶ X : the $T \times 6$ matrix of regressors; Hit : the vector of hits; $\hat{\beta}$: OLS estimate; u_t : regression error
 - ▶ DQ: explained sum of squares scaled by the null variance; 6 degrees of freedom = 6 regressors; large DQ = predictable hits
- Nests Kupiec and Christoffersen
 - ▶ $X_t = 1$ gives the score form $T(\hat{\pi} - \alpha)^2 / (\alpha(1 - \alpha))$; one lag gives Christoffersen
 - ▶ adds power against memory beyond lag 1 and hits that depend on the VaR level

All VaR Tests at a Glance



- p -values of the Kupiec, conditional coverage, duration and DQ tests, VaR 1%; red: rejection, green: no rejection
- DQ rejects every S&P 500 model, including FHS ($p < 0.001$) and GARCH-EVT ($p < 0.001$)
 - hits are predictable from the VaR level and lagged hits, a failure that counting breaches cannot detect
- Different tests examine different failures: read the pattern, not one p -value
 - the unconditional models fail the duration test almost everywhere
 - GARCH-EVT passes coverage and durations on BET, Bitcoin and EUR/RON

Size and Power of VaR Backtests: a Monte Carlo

- DGP (data generating process, the simulated truth): GARCH(1,1)-t fitted to the S&P 500 1990–2026 ($a = 0.102$, $b = 0.892$, $\nu = 6.1$); VaR 1%; 1000 paths; rejection rates (%) at 5%

Forecaster	T	POF	Cond. cov.	Duration	DQ
Correct (size)	250	9.8	1.2	14.8	8.6
	1000	6.5	3.3	7.9	11.0
	5000	4.9	3.4	4.6	8.8
Normal-GARCH (power)	250	10.5	4.9	14.7	18.8
	1000	30.3	19.9	8.0	30.3
	5000	83.8	77.2	7.2	70.7
HS, 1000 days (power)	250	51.2	17.2	12.3	32.4
	1000	46.3	47.1	68.5	73.8
	5000	57.1	86.4	100.0	99.5

Size and Power of VaR Backtests: Reading the Table

- The three forecasters
 - ▶ Correct: true parameters and true σ_t (rejection rate = size)
 - ▶ Normal-GARCH: true σ_t , Normal tail (rejection rate = power)
 - ▶ HS, 1000 days: no volatility dynamics (rejection rate = power)
- Conventions
 - ▶ an undefined duration p (fewer than 3 durations, 28% of correct samples at $T = 250$) counts as non-rejection
 - ▶ Monte Carlo standard error: 0.7 points at 5%, 1.6 at 50%
- The bold cells
 - ▶ in one year a Normal tail on the right volatility is almost invisible
 - ▶ HS clustering is seldom detected by the duration test at $T = 250$
 - ▶ DQ rejects a correct model more often than 5% even at $T = 5000$: simulate its critical values

Backtests Under Estimation Risk

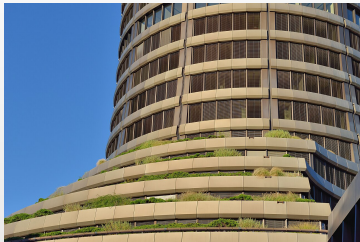
- ▣ Forecasts use the parameter estimate $\hat{\theta}$ obtained on R days; the test uses P out-of-sample days; the hits $\hat{I}_t = 1\{L_t > \text{VaR}_t(\hat{\theta})\}$ are not Bernoulli(α) even under a correct model
- ▣ The correction depends on the forecasting scheme (fixed, rolling, recursive) and grows with P/R (Escanciano & Olmo (2010); expansion in the [Appendix](#))
- ▣ Fixed-scheme illustration: GARCH(1,1)-t with the S&P 500 estimates, $R = 1000$, 400 paths: Kupiec rejects a **correct** model in 27.5% of samples at $P = 5000$, against 4.0% with the true parameters
- ▣ This chapter refits on a rolling window ($R = 1000$, every 20 days): the fixed-scheme number shows the mechanism, not the size of our tests
- ▣ Remedy: reproduce the estimation inside the calibration, e.g. a parametric bootstrap that re-estimates the model on every simulated path (Seminar 8, B9)

The Traffic Light: a Binomial Count Rule

Exceptions	$P(X \leq x)$	Zone
0	8.11%	green
2	54.32%	green
4	89.22%	green
5	95.88%	yellow
7	99.60%	yellow
9	99.97%	yellow
10	99.99%	red

- ▣ Banks with internal models backtest daily VaR 1% (BCBS (1996))
 - ▶ against actual P&L (profit and loss) over the last 250 days
- ▣ $X \sim \text{Binomial}(250, 0.01)$: exceptions of a correct VaR 1% in 250 days
- ▣ Yellow starts where $P(X \leq x) \geq 95\%$, red where $P(X \leq x) \geq 99.99\%$
- ▣ Only the **number** of exceptions counts, not their size or timing: simple, automatic, hard to game

Backtesting Rules in 1996 and 2019



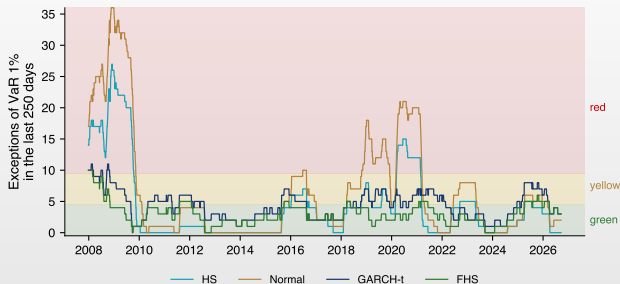
BIS tower, Basel: seat of the BCBS secretariat

Photo: Taxiarchos228 (2014); Free Art License;

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- **BCBS (1996):** capital on VaR 1% over 10 days
 - ▶ multiplier 3, plus 0.40–0.85 in the yellow zone and plus 1 in the red zone
- **BCBS (2019) (FRTB):** capital on ES 2.5%, backtesting still on VaR
 - ▶ bank-wide: VaR 1%, 250 days; multiplier 1.50 (green, 0–4), 1.70–1.92 (yellow, 5–9), 2.00 (red, 10 or more)
 - ▶ trading desk: more than 12 exceptions of VaR 1% or 30 of VaR 2.5% in 12 months move the desk to the standardised approach
 - ▶ IMA approval also requires a P&L attribution test (not covered here)
- VaR counts stay because they are simple
 - ▶ ES is not elicitable (section “Elicitability and Scoring Functions”)
 - ▶ yet ES can be backtested (section “Backtesting Expected Shortfall”)

The Traffic Light Through Time



- VaR 1% exceptions in the last 250 days, S&P 500; bands: green, yellow, red zones
- Share of days in red: Normal 22.3%, HS 14.7%, GARCH-t 2.0%, FHS 0.8%
- Peak: HS 27, Normal 37 exceptions
 - ▶ in the 250 days ending 01.12.2008; GARCH-t is yellow 46.6% of the time
- Overlapping windows: the penalty lasts a year
 - ▶ after a crisis, when capital is already strained

 MFM_ch8_traffic_light

FRTB Desk Check on the Last 250 Days (Limits 12 and 30)

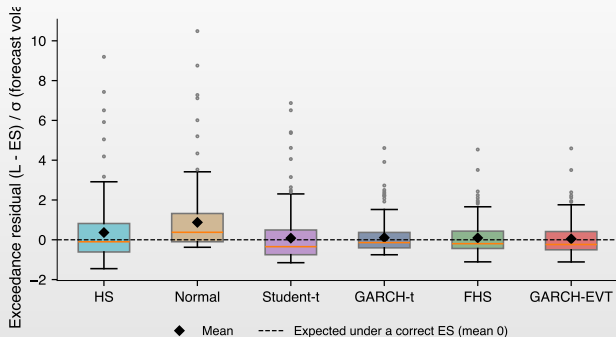
Model	S&P 500	BET	Bitcoin	EUR/RON
HS	0 / 2	4 / 7	3 / 4	1 / 5
Normal	2 / 2	6 / 9	3 / 5	1 / 4
Student-t	0 / 2	4 / 9	1 / 4	5 / 8
GARCH-t	3 / 6	4 / 9	3 / 5	4 / 5
FHS	3 / 4	4 / 6	4 / 5	1 / 4
GARCH-EVT	3 / 3	2 / 7	4 / 5	1 / 4

- Cells: exceptions of VaR 1% / VaR 2.5%
 - ▶ last 250 days of each series, to 18.09.2026
 - ▶ every model is well inside the limits
- Illustration only
 - ▶ FRTB applies the limits to desk P&L (actual and hypothetical), not to index returns
- The last year was calm
 - ▶ a green light today says little about the next crisis

McNeil and Frey (2000): Exceedance Residuals

- ES is an **average** beyond VaR: the information sits on about αT tail days (124 for the S&P 500 at $\alpha = 2.5\%$; 6.25 in one year, so a tail-mean standard error of order $1/\sqrt{\alpha T} \approx 0.40$)
- On each breach day of VaR 2.5%: $e_t = (L_t - \text{ES}_t)/\sigma_t$, with $\sigma_t > 0$ the forecast volatility, known at $t - 1$
- If VaR **and** ES are correct: $E[e_t \mid L_t > \text{VaR}_t, \mathcal{F}_{t-1}] = 0$; a wrong VaR changes the conditioning event
- H_0 : $E[e_t] = 0$ against H_1 : $E[e_t] > 0$ (ES underestimated); one-sided t -statistic
- Few, fat-tailed residuals: p -value from an i.i.d. bootstrap of the centred residuals (McNeil & Frey (2000)), which assumes independent residuals; otherwise a block bootstrap

Exceedance Residuals on the S&P 500



- Distribution of $(L_t - ES_t) / \sigma_t$ on breach days of VaR 2.5%; diamonds: means
- Means: Normal 0.875, HS 0.363, GARCH-t 0.106, FHS 0.093, GARCH-EVT 0.045
- Bootstrap p
 - ▶ Normal <0.001, HS <0.001, GARCH-t 0.023
 - ▶ FHS 0.096, GARCH-EVT 0.246
- Normal: tail mean too low by almost one volatility unit
 - ▶ the outliers (up to 10 units) decide capital adequacy

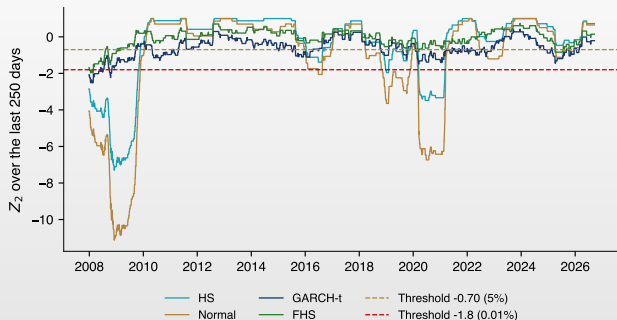
Acerbi and Szekely (2014): Testing ES Directly

- Breach indicator at the ES level: $I_t = 1\{L_t > \text{VaR}_t^\alpha\}$, $N_T = \sum_t I_t$ (the number of breaches in T days), $\alpha = 0.025$; ES_t : the ES forecast for day t
- $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{\text{ES}_t}$: is the average tail loss as large as forecast? (conditional on breaches)
- $Z_2 = 1 - \frac{1}{T_\alpha} \sum_t \frac{L_t I_t}{\text{ES}_t}$: tests frequency and size together
- Under H_0 (correct forecast distribution) $E[Z_2] = 0$; underestimated risk gives $Z_2 < 0$ (Acerbi & Szekely (2014))
- For $T = 250$ and many distributions, the 5% and 0.01% critical values are close to -0.70 and -1.8 : reference thresholds for an ES traffic light

Getting p -Values by Simulation

- The null distribution of Z_2 depends on the forecast distributions F_t (the distribution function of L_t forecast by the model on day $t - 1$)
- Algorithm (repeat $M = 2000$ times)
 - ▶ For each day draw $U_t \sim \text{Uniform}(0,1)$ and set $L_t^* = F_t^{-1}(U_t)$, a loss drawn from the forecast (F_t^{-1} : the quantile function)
 - ▶ Compute Z_2^* from the simulated losses with the same VaR and ES forecasts
- p -value = share of simulated Z_2^* at or below the observed Z_2
- Only tail draws matter: for HS and FHS we store the 25 largest window losses of each day
- S&P 500, full sample: $Z_2 = -0.618$ for GARCH-t ($p < 0.001$) versus -0.091 for FHS ($p = 0.172$); critical value about -0.152

Z_2 Over Rolling 250-Day Windows



- Z_2 of ES 2.5%, S&P 500, last 250 days
 - ▶ reference thresholds -0.70 and -1.8
- Minimum: Normal -11.13 , HS -7.30
 - ▶ window ending 01.12.2008; GARCH-t -2.52 , FHS -1.97
- Windows below -1.8
 - ▶ Normal 21.9%, HS 15.8%, GARCH-t 3.0%, FHS 0.7%
 - ▶ GARCH-t below -0.70 in 39.6% of windows
- Daily checks are repeated tests
 - ▶ false alarms accumulate (sequential monitoring, [Appendix](#))

 MFM_ch8_es_backtests

Du and Escanciano (2017): Cumulative Violations

- ▣ PIT (probability integral transform): $u_t = F_t(L_t)$, the forecast probability of a loss at most L_t ; uniform on $(0, 1)$ if the forecast distribution F_t is correct
- ▣ Cumulative violation: $H_t = \frac{1}{\alpha} (u_t - (1 - \alpha)) 1\{u_t > 1 - \alpha\}$: how deep into the tail the loss went
- ▣ Under H_0 : $E[H_t] = \alpha/2$ and $\text{Var}(H_t) = \alpha(1/3 - \alpha/4)$
- ▣ Unconditional test: $U = \sqrt{T} (\bar{H} - \alpha/2) / \sqrt{\alpha(1/3 - \alpha/4)} \xrightarrow{d} N(0, 1)$, with $\bar{H}$ the sample mean of H_t ; $U > 1.645$: losses go deeper into the tail than forecast (one-sided, 5%)
- ▣ Conditional test: Box–Pierce statistic on the first 5 autocorrelations of H_t , χ_5^2 (Du & Escanciano (2017))

All ES Tests at a Glance

	McNeil-Frey				Acerbi-Szekely Z2				Du-Escanciano U			
HS	<0.001	0.48	0.70	0.46	<0.001	0.13	0.86	1.00	<0.001	0.36	0.14	<0.001
Normal	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	0.97	<0.001	<0.001	<0.001	0.25
Student-t	0.17	0.01	1.00	0.47	0.002	0.003	0.62	0.92	<0.001	<0.001	0.93	0.05
GARCH-t	0.02	0.04	0.50	0.41	<0.001	<0.001	0.05	0.63	<0.001	<0.001	0.13	0.86
FHS	0.10	0.63	0.60	0.75	0.17	0.19	0.35	0.97	0.09	0.46	0.95	0.09
GARCH-EVT	0.25	0.57	0.29	0.69	0.09	0.21	0.58	0.98	0.24	0.54	0.70	0.11
	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON	S&P 500	BET	Bitcoin	EUR/RON

- p -values for ES 2.5%: McNeil-Frey (bootstrap), Z_2 (simulation), Du-Escanciano U ; the Normal distribution is rejected by all three on S&P 500, BET and Bitcoin; FHS and GARCH-EVT pass all three everywhere
- Different nulls and inputs, different decisions: Student-t on the S&P 500 has McNeil-Frey $p = 0.173$ but $Z_2 p = 0.002$; these are calibration tests, not rankings
- GARCH-t fails Z_2 on equities ($p < 0.001$, < 0.001) while FHS passes; EUR/RON: most models over-predict ES ($Z_2 > 0$)

Multinomial VaR Backtests: an Implicit ES Test

- ES 2.5% averages the quantiles beyond VaR 2.5%: test several VaR levels at once (Kratz, Lok & McNeil (2018))
- $N = 4$ tail levels $\alpha_j = 2.5\% (1 - (j - 1)/4)$: 2.5%, 1.875%, 1.25%, 0.625%; each day falls in one of 5 cells via the PIT u_t
- Pearson statistic $S = \sum_{j=0}^4 (O_j - Tp_j)^2 / (Tp_j) \xrightarrow{d} \chi_4^2$: O_j observed and Tp_j expected days in cell j , cell probabilities $p_0 = 97.5\%$, $p_1 = \dots = p_4 = 0.625\%$; large S = counts far from expected

p -value	S&P 500	BET	Bitcoin	EUR/RON
HS	<0.001	0.725	0.564	0.004
Normal	<0.001	<0.001	<0.001	<0.001
Student-t	<0.001	0.024	0.015	0.172
GARCH-t	<0.001	<0.001	0.227	0.568
FHS	0.008	0.846	0.413	0.392
GARCH-EVT	0.287	0.864	0.969	0.240

- Only GARCH-EVT passes in all four markets; S&P 500 GARCH-t cells (4767, 36, 46, 50, 60) against 31 expected in each tail cell: too many days in every tail cell, most in the deepest

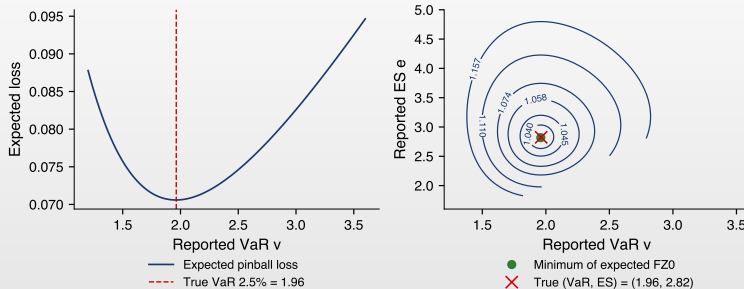
Testing Is Not Ranking: Consistent Scores

- A backtest asks whether a model is **acceptable** (several can pass, all can fail); comparative backtesting asks which is **better** (Nolde & Ziegel (2017))
- Ranking needs a **scoring function** $S(x, L)$, the loss of reporting x when L occurs; compare $\bar{S} = \frac{1}{T} \sum_t S(x_t, L_t)$, lower is better
- S is **consistent** for $\theta(F)$, and θ is **elicitable**, if $\theta(F) = \arg \min_x E_F[S(x, L)]$ (Gneiting (2011))
 - ▶ the truth minimises the expected score
 - ▶ $\theta(F)$: the target functional (e.g. VaR) of the loss distribution F ; E_F : expectation when $L \sim F$
- VaR: pinball loss $S(v, L) = (1\{L > v\} - \alpha)(L - v)$, v : the VaR forecast; for $E|L| < \infty$ and density $f > 0$ at the quantile
 - ▶ $\frac{\partial}{\partial v} E_F[S(v, L)] = F(v) - (1 - \alpha) = 0 \Rightarrow v = F^{-1}(1 - \alpha) = \text{VaR}_\alpha$, and $\partial^2 / \partial v^2 = f(v) > 0$
- ES is **not** elicitable on its own: no scoring function of the reported ES forecast e alone is minimised by the true ES (Gneiting (2011))

VaR and ES Together: Fissler and Ziegel (2016)

- The pair (VaR, ES) is jointly elicitable: [Fissler & Ziegel \(2016\)](#) characterise the whole family of consistent scoring functions
- FZ0 loss ([Patton, Ziegel & Chen \(2019\)](#)), with losses positive, $0 < v \leq e$, $E|L| < \infty$ and a true ES above 0
 - ▶
$$\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$$
 - ▶ v, e : the VaR and ES forecasts (positive losses); L : the realised loss
- First term: penalty for breaches, scaled by ES; last terms: penalty for reporting a large ES
- Differences of FZ0 losses do not depend on the units of the losses, so they can be compared across assets
- Lower average FZ0 = better joint VaR and ES forecast

Elicitability in Pictures

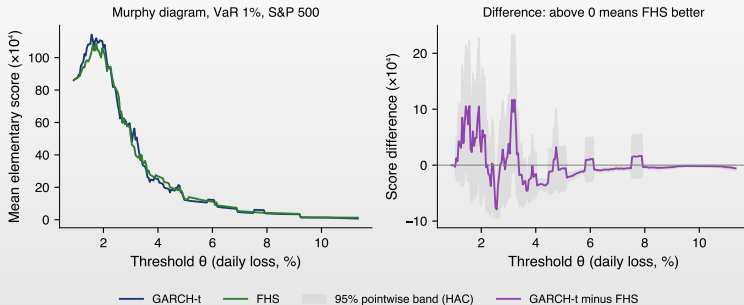


- Student-t losses with $\nu = 4$ and unit variance, $\alpha = 2.5\%$; expected losses computed on 2 million draws
- Left: the expected pinball loss is lowest at the true VaR 2.5%, 1.96; right: the expected FZ0 loss is lowest at (1.96, 2.82), the true (VaR, ES) up to the grid
- A forecaster who knows the truth has no incentive to report anything else: this is what makes ranking by average score meaningful

Is the Ranking Robust to the Score? Elementary Scores

- For a quantile there are infinitely many consistent scores: $S(v, L) = (1\{L \leq v\} - \tau)(g(v) - g(L))$, any increasing function g , $\tau = 1 - \alpha$ (the quantile level of the loss)
- Each is a mixture of **elementary scores** (Ehm, Gneiting, Jordan & Krüger (2016))
 - ▶ $S_\theta(v, L) = (1\{L < v\} - \tau)(1\{\theta < v\} - 1\{\theta < L\})$, $\theta \in \mathbb{R}$; θ : a threshold on the loss scale; S_θ penalises a forecast only when v and L fall on different sides of θ ; the pinball loss is $\int S_\theta d\theta$
- Forecast A beats B under **every** consistent score iff $\bar{S}_\theta(A) \leq \bar{S}_\theta(B)$ for all θ , with $\bar{S}_\theta$ the average elementary score over the T days
- **Murphy diagram**: plot $\bar{S}_\theta$ against θ ; crossing curves mean the ranking depends on the chosen score

Murphy Diagram: GARCH-t versus FHS

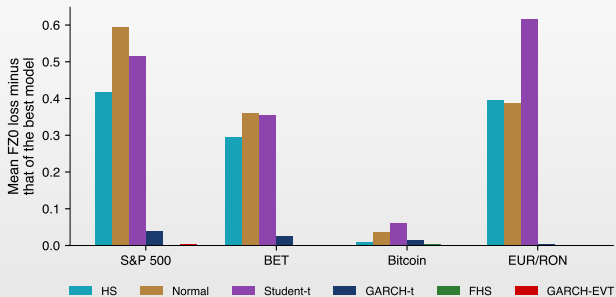


- S&P 500, VaR 1%, 2007–2026; θ from 0.9% to 11.3%; pinball ($\times 10^4$): GARCH-t 3.849, FHS 3.849, a tie under the usual score
- Below $\theta = 5\%$ the curves cross: FHS better on 50% of thresholds, band containing zero on 82%; above 5% GARCH-t lower on 89%: no dominance
- The bands are pointwise, not a simultaneous test of dominance: state the score before looking at the results

The Diebold–Mariano Test on FZ0 Losses

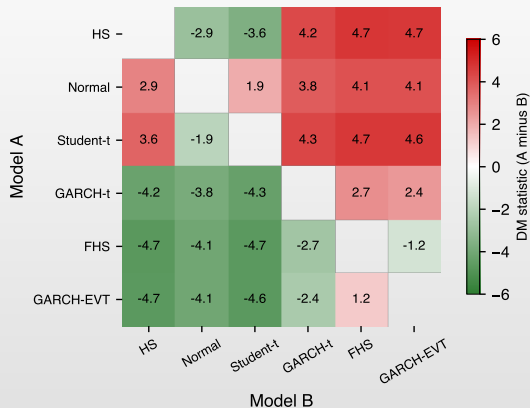
- Loss differential of models A and B : $d_t = \text{FZ0}_{A,t} - \text{FZ0}_{B,t}$
- H_0 : $E[d_t] = 0$ (equal predictive accuracy), [Diebold & Mariano \(1995\)](#); the statistic is the mean differential $\bar{d}$ divided by its standard error
- $$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d_t)/T}} \xrightarrow{d} N(0, 1)$$
- $\widehat{\text{LRV}}$: long-run variance with Newey–West weights ([Newey & West \(1987\)](#)), 9 lags; tail losses cluster, so d_t is autocorrelated
- $\text{DM} > 0$: model A has the larger average loss (worse); $|\text{DM}| > 1.96$: significant at 5%

Average FZ0 Loss by Model



- Mean FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts minus that of the best model in each market
- Lowest: FHS on the S&P 500; GARCH-EVT on BET, Bitcoin and EUR/RON
- Unconditional models: 0.3–0.6 FZ0 units worse
 - ▶ on S&P 500, BET and EUR/RON; on Bitcoin the largest gap is 0.062
- GARCH-t trails FHS on the S&P 500 by 0.039
 - ▶ the size of a gap does not tell whether the data separate the models: we test it

Pairwise Diebold–Mariano Tests, S&P 500



- DM of model *A* (row) minus *B* (column) on FZ0 losses; red: *A* worse, green: *A* better
- Against FHS
 - ▶ HS DM = 4.70, GARCH-t DM = 2.65: both significantly worse
 - ▶ GARCH-EVT DM = 1.24: indistinguishable
- Normal minus HS: DM = 2.88: a slow empirical quantile still beats a thin tail
- Fifteen pairwise tests: a multiple-testing problem
 - ▶ which models survive **jointly**? (MCS, below)

Comparing Forecasting Methods: Giacomini–White

- DM targets the population accuracy of **models**
 - ▶ with rolling 1000-day windows and refits we compare **forecasting methods**; estimation error never vanishes
- Giacomini & White (2006): $H_0: E[d_t | \mathcal{F}_{t-1}] = 0$ for the score differential $d_t = FZ0_{A,t} - FZ0_{B,t}$; valid for nested models and finite windows
- Test function $h_{t-1} = (1, d_{t-1})'$: information known at $t - 1$ that might predict which method wins; $Z_t = h_{t-1} d_t$
 - ▶ $GW = T \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \xrightarrow{d} \chi^2_2$, $\hat{\Omega} = T^{-1} \sum_t Z_t Z_t'$ (one step ahead); $\bar{Z}$: the mean of Z_t , $\hat{\Omega}$: its estimated covariance; 2 degrees of freedom = 2 instruments; $h_{t-1} = 1$ gives the DM form
- S&P 500 against FHS: GARCH-EVT $GW = 7.38$ ($p = 0.025$), GARCH-t 7.59 ($p = 0.022$), HS 53.38 ($p < 0.001$)
- GARCH-EVT, a tie under DM, is rejected conditionally
 - ▶ equal conditional predictive ability fails for these instruments
 - ▶ what drives the rejection needs a separate test

Three-Zone Comparative Backtesting (Nolde and Ziegel)

- Internal model A against a standard model S (here HS), mean FZ0 loss; two one-sided tests at 5% (Nolde & Ziegel (2017))
- H_0^- : A at least as good as S ; H_0^+ : A at most as good. **Green**: reject H_0^+ ; **red**: reject H_0^- ; **yellow**: neither
- The burden of proof is reversed: the bank must show that its model is better, not the supervisor that it is worse

DM (A minus HS)	S&P 500	BET	Bitcoin	EUR/RON
GARCH-t	−4.23	−5.18	0.09	−6.66
FHS	−4.70	−5.49	−0.17	−6.94
GARCH-EVT	−4.68	−5.53	−0.25	−6.92

- Colour = zone. Green everywhere except Bitcoin, where every model is yellow: with its noisy tail, no model can prove it beats HS

The Model Confidence Set

- Model confidence set, MCS ([Hansen, Lunde & Nason \(2011\)](#))
 - ▶ contains the best models with asymptotic probability at least $1 - \alpha_{MCS}$
 - ▶ $\alpha_{MCS} = 10\%$ below; under the method's regularity conditions
- Algorithm
 - ▶ Test H_0 : all models in the current set have equal expected loss
 - ▶ If rejected, eliminate the model with the largest standardised average loss and repeat
 - ▶ Critical values from a block bootstrap (blocks of 10 days, 1000 replications)
- MCS p -value of a model: the largest test p -value up to its elimination; keep models with $p \geq 0.10$
- A large MCS is an honest answer: the data do not separate the models

Which Models Survive?

MCS p-values (FZ0 loss); models with $p \geq 0.10$ are in the 90% MCS

	S&P 500	BET	Bitcoin	EUR/RON
HS	<0.001	<0.001	0.87	<0.001
Normal	<0.001	<0.001	0.63	<0.001
Student-t	<0.001	<0.001	0.30	<0.001
GARCH-t	0.009	0.09	0.84	0.90
FHS	1.00	0.90	0.87	0.99
GARCH-EVT	0.21	1.00	1.00	1.00

- MCS p -values, FZ0 loss of (VaR 2.5%, ES 2.5%)
 - ▶ models with $p \geq 0.10$ form the 90% MCS
- S&P 500: FHS and GARCH-EVT
 - ▶ GARCH-EVT MCS p 0.21; GARCH-t out (p 0.01)
- BET: FHS and GARCH-EVT
 - ▶ GARCH-t just out (p 0.09)
- EUR/RON: the three conditional models
- Bitcoin: all six models stay
 - ▶ the data cannot separate them; non-rejection is not proof of equal performance

Split Conformal Quantiles: a Distribution-Free Promise

- **Conformal prediction** (Lei et al. (2018), Angelopoulos & Bates (2021))
 - ▶ the models so far assume a distribution (Normal, t, GPD) or a stable history (HS)
 - ▶ conformal prediction turns any forecast into a quantile with a coverage guarantee
- Score $s_t = L_t / \hat{\sigma}_t$, $\hat{\sigma}_t$ the standard deviation of the previous 250 returns; calibration set: the last $n = 250$ scores, sorted $s_{(1)} \leq \dots \leq s_{(n)}$
- Conformal VaR: $\text{VaR}_t = \hat{\sigma}_t s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$ ($\lceil x \rceil$: smallest integer $\geq x$): today's volatility times the k -th smallest past score; $n = 250$, $\alpha = 0.01$: $k = 249$, the second largest score
- Guarantee if the $n+1$ scores are **exchangeable** (joint law invariant to reordering):
 $1 - \alpha \leq P(L_t \leq \text{VaR}_t) \leq 1 - \alpha + \frac{1}{n+1}$ (no ties)
- Here the guarantee is only approximate (extensions: [Appendix](#))
 - ▶ volatility clustering breaks exchangeability; $\hat{\sigma}_t$ comes from overlapping rolling windows
 - ▶ genuine split conformal fits the score once, on a separate sample

Adaptive Conformal Inference

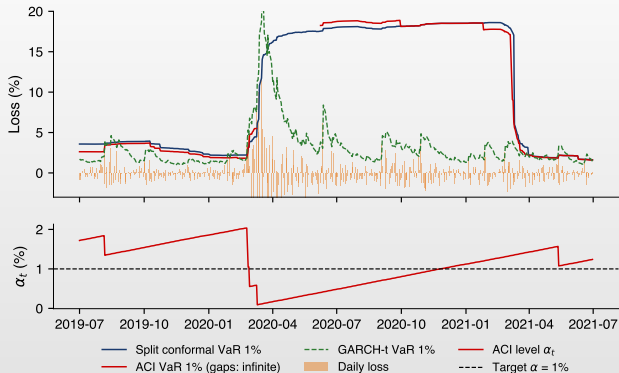


Emmanuel Candès, co-author of adaptive conformal inference

Photo: Cmiche167 (2026); CC BY-SA 4.0;
Wikimedia Commons

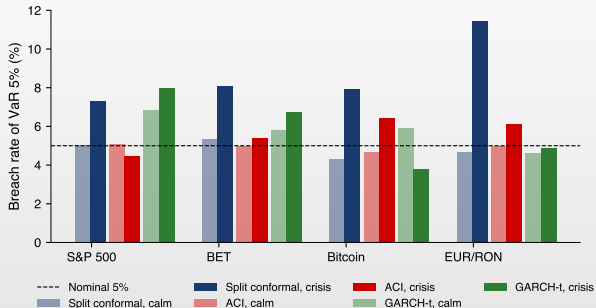
- Gibbs & Candès (2021): the working level is updated after each day
 - ▶ $\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t)$, $\text{err}_t = 1\{L_t > \text{VaR}_t\}$
 - ▶ α_t : level of day t 's conformal quantile; $\gamma > 0$: step size; α_1 : starting level
- After a breach α_t falls (VaR rises); after calm days it slowly rises again; we use $\gamma = 0.005$
- Guarantee for any sequence: $\left| \frac{1}{T} \sum_t \text{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$, a long-run **average**, not each day or each crisis
- If $\alpha_t < 1/(n+1)$, the required quantile does not exist: the VaR becomes infinite

Conformal VaR Through COVID-19



- S&P 500, July 2019–June 2021, VaR 1%
 - ▶ top: split conformal, ACI and GARCH-t; bottom: ACI level α_t
- 20.02–30.04.2020: 50 days, 0.5 breaches expected
 - ▶ split conformal 6, ACI 4 (8% of days)
- ACI set $\text{VaR} = \infty$ on 37 of these days
 - ▶ no further breaches, but an infinite VaR carries no information
- Split conformal stays near 18% for a year
 - ▶ March 2020 stays in its calibration set; GARCH-t adapts within days

Coverage in Calm and Crisis Periods



- VaR 5%; crisis day: known in advance
 - ▶ previous 20-day volatility above the 90th percentile of the previous 1000 days
- Split conformal breach rates
 - ▶ all days (S&P 500, BET, Bitcoin, EUR/RON): 5.34%, 5.63%, 4.58%, 5.06%
 - ▶ crisis days: 7.29%, 8.08%, 7.95%, 11.43%
- ACI: equity crisis rates near 5%
 - ▶ S&P 500 4.49%, BET 5.38%
 - ▶ for VaR 1% only by an infinite VaR on 13.4% of S&P 500 days
 - ▶ marginal coverage is not crisis coverage

Case Study (1/5): Can ES Be Backtested Using Only ES Forecasts?

- Bayer & Dimitriadis (2022): can ES forecasts be backtested with nothing but the realised returns and the ES forecasts themselves?
- Why it matters: FRTB puts capital on ES 2.5% (section on the regulator's view)
 - ▶ the ES tests of the section “Backtesting Expected Shortfall” also need VaR forecasts, the volatility or the whole forecast distribution
 - ▶ supervisors do not necessarily receive these inputs
 - ▶ a rejection of such a joint test does not say which input is wrong (paper, Sec. 1.1, Definition 1.1)
- Idea: a Mincer–Zarnowitz regression of the returns on the ES forecasts, H_0 : intercept 0, slope 1 (a correct forecast needs no correction)
 - ▶ ES alone is not elicitable (section “Elicitability and Scoring Functions”), but (VaR, ES) is jointly elicitable (Fissler & Ziegel (2016)), so a joint quantile–ES regression can be estimated (Patton, Ziegel & Chen (2019))

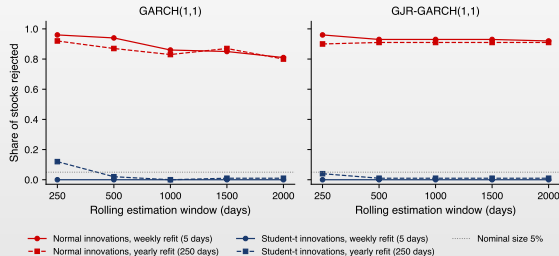
Case Study (2/5): the ES Regression

- Returns convention: $\hat{e}_t = -\widehat{\text{ES}}_t < 0$, the ES forecast on the return scale, $\tau = 2.5\%$ (the paper's symbol for our α)
 - ▶ ES equation (Eq. 1.4–1.6): $r_t = \gamma_1 + \gamma_2 \hat{e}_t + u_t^e$, $\text{ES}_\tau(u_t^e | \mathcal{F}_{t-1}) = 0$
 - ▶ γ_1 : intercept, γ_2 : slope; u_t^e : error whose conditional ES is zero, so $\gamma_1 + \gamma_2 \hat{e}_t$ is the true conditional ES; correct forecasts: $\gamma_1 = 0$, $\gamma_2 = 1$
- ESR (expected shortfall regression) also needs a quantile equation, because ES is estimated jointly with VaR
 - ▶ **Auxiliary ESR**: quantile equation on $(1, \hat{v}_t)$, needs VaR forecasts $\hat{v}_t$; **Strict ESR**: quantile equation on $(1, \hat{e}_t)$, ES forecasts only
 - ▶ **Intercept ESR** (Eq. 1.14–1.15): $r_t - \hat{e}_t = \gamma_1 + u_t^e$, slope fixed at 1; one-sided t -test of $H_0: \gamma_1 \geq 0$ against $\gamma_1 < 0$ (ES underestimated)

Case Study (3/5): Estimation, Covariance and Monte Carlo Evidence

- M-estimator minimising an FZ0-type loss (paper, Eq. 1.16–1.18)
 - ▶ asymptotics under misspecification (paper, Theorem 1.4)
 - ▶ R packages [esback](#) and [esreg](#)
- Two assumptions to keep in mind
 - ▶ Strict and Intercept ESR: the quantile equation is correct only if VaR is an affine function of ES (e.g. pure scale models) (paper, Sec. 1.2)
 - ▶ The misspecification-robust sandwich covariance assumes a location-scale model for the conditional distribution (paper, Sec. 1.5)
- Monte Carlo design (paper, Sec. 2)
 - ▶ 10 000 replications, T from 250 to 5 000
 - ▶ EGARCH-t calibrated to the S&P 500, AR(1)-GARCH(1,1), two Student-t GAS models
 - ▶ competitors: [McNeil & Frey \(2000\)](#) and the calibration tests of [Nolde & Ziegel \(2017\)](#)
- Size at 5%, EGARCH-t, $T = 250$ and $T = 1\,000$ (Table 1)
 - ▶ Strict ESR: 0.09 and 0.05 with the robust covariance, 0.24 and 0.11 with the classical one
 - ▶ simple conditional calibration test: 0.29 and 0.14
- Power (Fig. 3): ESR detects what residual tests miss
 - ▶ exceedance-residual tests barely react to wrong volatility dynamics or a wrong level $\tilde{\tau} \neq \tau$
 - ▶ a bank could report ES for a milder level and pass the older tests

Case Study (4/5): Tails Matter More Than Refits



□ Bayer & Dimitriadis (2022), Table 3: 200 largest S&P 500 stocks, 2010–2019

- ▶ share with ES 2.5% rejected by the one-sided Intercept ESR at 5%
- ▶ windows of 250–2 000 days; refits every 5 (solid) or 250 days (dashed)

□ Normal innovations: 0.80–0.96 rejected

- ▶ in all 50 combinations (window $\times$ refit $\times$ GARCH/GJR-GARCH)

□ Student-t: at most 0.12

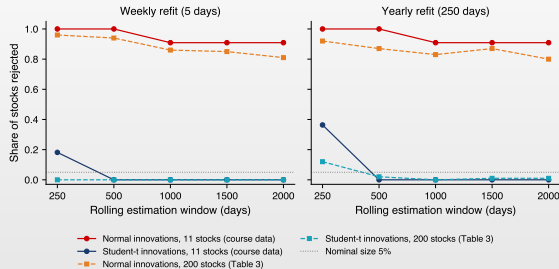
- ▶ up to 0.12 with 250-day windows, at most 0.01 from 1 000 days

□ Fat tails and a long window matter most

- ▶ the refit frequency matters only with short windows

 MFM_ch8_esr_case

Case Study (5/5): Table 3 Replicated on the Course Data



- Solid: 11 large S&P 500 stocks with prices from 2002
 - ▶ January 2010 – August 2019 (2 432 days); dashed: Bayer & Dimitriadis (2022), Table 3
 - ▶ GARCH(1,1) with a constant mean (the paper does not state the mean equation)
 - ▶ 25 window–refit combinations; Intercept ESR with `esback`
 - ▶ each rejected stock adds $1/11 \approx 0.09$ to a share
- Normal: 0.82–1.00 rejected (paper 0.80–0.96)
 - ▶ Student-t: at most 0.00 from 1 000 days (paper 0.01)
 - ▶ but 0.09–0.55 at 250 days (paper 0.00–0.12)
- The central result replicates; short-window shares are higher

An Open Question: Crisis-Conditional Coverage

- Distribution-free VaR guarantees are **marginal**: split conformal needs exchangeability, ACI controls only the long-run average (section “Conformal Prediction for Tail Risk”)
- Formal question, with a crisis flag $C_t \in \mathcal{F}_{t-1}$ known ex ante ($C_t = 1$ on a crisis day, 0 otherwise): is the breach rate on crisis days still α ?
 - ▶ $H_0: P(L_t > \text{VaR}_t \mid C_t = 1) = \alpha$ against $H_1: P(L_t > \text{VaR}_t \mid C_t = 1) \neq \alpha$
 - ▶ Falsified by a crisis breach rate outside the Binomial acceptance region, after correcting for dependent breaches and for multiplicity
- One pass of the discovery loop with an AI assistant (an LLM such as Claude, ChatGPT or Gemini)
 - ▶ literature with DOIs checked on Crossref; replicate a known number first
 - ▶ three rival explanations: stale scale, changing tail shape, regime-selection artefact
 - ▶ pre-register the tests; ask the assistant for a hostile referee report
 - ▶ Reusable prompts and a research checklist: [▶ Appendix](#)
- Stakes: capital is needed exactly on the days when marginal guarantees say least

Mini-Case: Does a GARCH Scale Repair Crisis Coverage?

- Assistant proposal: the standardised loss $s_t = (L_t + \mu_t)/\sigma_t$ as score
 - ▶ μ_t, σ_t : GARCH-t forecasts of the mean and volatility
 - ▶ H_0 : crisis breach rate of VaR 5% = 5%; exact Binomial test, Holm correction across the four markets

Market	Crisis days	Split conformal	Holm p	GARCH-scaled	Holm p
S&P 500	613	7.01%	0.052	2.77%	0.037
BET	506	8.10%	0.009	3.95%	0.309
Bitcoin	150	8.00%	0.092	1.33%	0.111
EUR/RON	245	11.43%	<0.001	2.45%	0.153

 ai_discovery_case.py

Mini-Case: Reading the Result

- Sample change first
 - ▶ the GARCH-scaled scores need 250 calibration days
 - ▶ common S&P 500 sample: 4 709 days, 613 crisis days (full sample of the conformal section: 713)
 - ▶ split conformal crisis rate: 7.01% (full sample: 7.29%)
- The GARCH scale pushes every crisis rate below 5%
 - ▶ Holm rejects only on the S&P 500 (2.77%)
- Elsewhere: non-rejection, not proof of coverage
 - ▶ Binomial p -values ignore clustered breaches
 - ▶ the crisis samples are small

Which Model Would You Provisionally Select?



National Bank of Romania, Bucharest

Photo: Britchi Mirela (2018); CC BY-SA 4.0;

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▣ Calibration

- ▶ only FHS and GARCH-EVT pass Z_2 at 5% and stay in the 90% MCS in all four markets (scorecard: [▶ Appendix](#))

▣ Comparison

- ▶ lowest FZ0: FHS on the S&P 500, GARCH-EVT elsewhere; the two do not differ significantly
- ▶ on Bitcoin no model can be separated from HS

▣ Unresolved

- ▶ DQ rejects every S&P 500 model; crisis-conditional coverage fails; one year of data has little power

▣ Provisional choice

- ▶ a GARCH filter with a flexible tail, the other as challenger, re-examined as evidence accrues
- ▶ a Romanian portfolio (BET and EUR/RON): Seminar 8, C1

Check Yourself, and Next: Chapter 9

Retrieval questions

- Why does the rule " $LR_{uc} > 3.84$ " reject a correct VaR 1% in about one year in ten?
- Why can a model with the right number of breaches still fail?
- Why can ES be backtested but not ranked on its own?
- What does ACI guarantee?
- What does ACI not guarantee?

Chapter 9: realised volatility

- Measure daily variance from intraday returns
- Evaluate volatility forecasts with robust scores (QLIKE) and the MCS

Appendix: Further Reading and Resources

- ▣ Key papers
 - ▶ Fissler & Ziegel (2016): VaR and ES are jointly elicitable, the basis of the FZ0 loss
 - ▶ Nolde & Ziegel (2017): comparative backtests and the traffic light for VaR and ES
 - ▶ Acerbi & Szekely (2014): model-free ES backtests, including Z_2
- ▣ Entropy-based VaR and ES forecasts: Pele, Lazar & Dufour (2017), Pele & Mazurencu-Marinescu-Pele (2019), Pele, Lazar & Mazurencu-Marinescu-Pele (2019)
- ▣ When two ES forecasts can be told apart: Pele & Mazurencu-Marinescu-Pele (2026)
 - ▶ about one in five pairwise ES comparisons in global markets is precision-fragile
 - ▶ a consequence of the small effective tail sample αT
- ▣ Textbooks: Franke, Härdle & Hafner (2019) (VaR and backtesting chapters) and Härdle, Chen & Overbeck (2017)
- ▣ Quantinar: Machine learning in Financial Risk
- ▣ Code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_08; S&P 500, BET, Bitcoin: daily closes from the course data; EUR/RON: BNR reference rate

Appendix: Estimation Risk, the Fixed-Scheme Expansion

- Forecasts use $\hat{\theta}$ estimated on R days; the test uses P out-of-sample days
- Fixed scheme (Escanciano & Olmo (2010)): $\frac{1}{\sqrt{P}} \sum_t (\hat{I}_t - \alpha) = \frac{1}{\sqrt{P}} \sum_t (I_t - \alpha) + \sqrt{P/R} D' \sqrt{R} (\hat{\theta} - \theta_0) + o_p(1)$
 - ▶ First term: the centred hits with the true parameters θ_0 ; second term: the effect of the estimation error $\hat{\theta} - \theta_0$
 - ▶ $D = E [\partial P(L_t > \text{VaR}_t(\theta) \mid \mathcal{F}_{t-1}) / \partial \theta]$ at θ_0 , the sensitivity of the breach probability to the parameters; the second term vanishes only if $P/R \rightarrow 0$
 - ▶ The estimation term adds a variance of order P/R to $\sqrt{P}(\hat{\pi} - \alpha)$, so χ^2 critical values are too small when P/R is large
- Rolling and recursive schemes change the form of the correction; the expansion assumes smooth $\text{VaR}_t(\theta)$, a $\sqrt{R}$ -consistent $\hat{\theta}$ and stationarity
- Remedies: the corrected variance or subsampling of Escanciano & Olmo (2010), or a parametric bootstrap that re-estimates the model on every simulated path
- Fixed scheme, 400 paths, $R = 1000$: Kupiec rejection 11.5% / 12.8% / 27.5% at $P = 250$ / 1000 / 5000 with estimated parameters, 10.0% / 5.2% / 4.0% with the true ones; Monte Carlo standard error about 1.1 points at 5%

Appendix: Osband's Principle and Identification Functions

- ▣ Osband's principle ([Fissler & Ziegel \(2016\)](#)): consistent scores are built from an identification function V , with $E_F[V(x, L)] = 0$ only at the true value
- ▣ VaR: $V(v, L) = 1\{L \leq v\} - (1 - \alpha)$; the pinball loss is its integral in v
- ▣ (VaR, ES): $V_1 = \alpha - 1\{L > v\}$ and $V_2 = e - v - \frac{1}{\alpha}1\{L > v\}(L - v)$; their expectations vanish only at the true pair
- ▣ The level sets of ES are not convex under mixtures, so no identification function of e alone exists (Seminar 8, A9)
- ▣ FZ0 is the member of the FZ family whose differences are invariant to the scale of the losses ([Patton, Ziegel & Chen \(2019\)](#))

Appendix: Sequential Monitoring of Risk Forecasts

- A regulator checks the forecasts every day: repeating a fixed-sample test (e.g. rolling Z_2) inflates the probability of at least one false alarm
- Hoga & Demetrescu (2023): monitoring statistics for VaR and ES with boundary functions that keep the probability of any false alarm over the whole horizon below the nominal level
- Compared with the traffic light: evidence accumulates from the start of monitoring instead of a single 250-day snapshot
- The ES regression backtest (Bayer & Dimitriadis (2022)) of the case study can be monitored the same way; both address the one-shot design of the traffic light

Appendix: Conformal Prediction Beyond Exchangeability

- Barber, Candès, Ramdas & Tibshirani (2023): coverage without exchangeability
 - ▶ the coverage gap is bounded by weighted total-variation distances
 - ▶ between the score vector and the same vector with the test score swapped with calibration score i
- How large these distances are under volatility clustering is an empirical question
- Gibbs, Cherian & Candès (2025): exact coverage conditional on a pre-specified class of covariate shifts is attainable; which class captures a financial crisis is open
- Choosing the ACI step γ : Zaffran et al. (2022) aggregate ACI experts over a grid of γ values

Appendix: Case Study, Complete Key Results (1/2): Size and Power

Finding	Source	Reading
Size at nominal 5%, EGARCH-t: Strict ESR 0.09 ($T = 250$) and 0.05 ($T = 1\,000$) with the robust covariance; 0.24 and 0.11 with the classical one	Table 1	The robust covariance improves size control; at $T = 250$ the size is still 0.09
Strict and Auxiliary ESR sizes differ by at most 0.01 for every process and sample size, including AR(1)-GARCH with $\phi = 0.5$	Table 1	In these designs, dropping the VaR forecasts barely changes the size; power is compared separately (Fig. 2–3)
Simple conditional calibration test: size 0.29 at $T = 250$ and 0.14 at $T = 1\,000$ (EGARCH-t)	Table 1	A competitor with high raw power can owe it to over-rejection

Appendix: Case Study, Complete Key Results (2/2): Power and Stock Data

Finding	Source	Reading
Exceedance-residual tests barely react to wrong volatility dynamics or to forecasts for a wrong level $\tilde{\tau} \neq \tau$; ESR tests detect every misspecification	Fig. 3	A bank could report ES for a milder level and pass the older tests
Normal innovations: rejected for 80%–96% of the stocks in every window and refit design	Table 3	Normal tails understate ES for most stocks
Student-t: GARCH 0.12 with a 250-day window and yearly refit; at most 0.01 with 1 000 days or more, for GARCH and GJR-GARCH; leverage: from 500 days GJR-GARCH-t and GARCH-t differ by at most 0.01	Table 3	For these 200 stocks and 2010–2019: fat tails and windows of 1 000 days or more give rejection shares of at most 0.01

Appendix: Case Study, Limitations and Extensions

- Limits stated by the authors
 - ▶ Strict and Intercept ESR: the quantile equation is misspecified unless VaR is an affine function of ES (paper, Sec. 1.2); Auxiliary and Strict ESR are two-sided only (paper, Sec. 1.3)
 - ▶ The classical covariance over-rejects in small samples (paper, Sec. 2.1)
- Replication with the course data (Seminar 8, C3)
 - ▶ Run the Strict ESR (two-sided) and the Intercept ESR (one-sided) on the ES 2.5% forecasts of the six models in the four markets; compare with Z_2 and McNeil–Frey
 - ▶ Extend the Table 3 replication: GJR-GARCH, more stocks and the period after August 2019
- Open: a one-year supervisory window
 - ▶ At $T = 250$ the two-sided Intercept ESR has size 0.04 with the robust covariance and 0.15 with the classical one (Table 1, EGARCH-t); the one-sided size is reported only for $T = 2500$ (0.02, Table 2)
 - ▶ What is the one-sided size at one year?
 - ▶ Can a bootstrap of the t -statistic hold it at 5% without losing the power of Fig. 4?

Appendix: Scorecard Across Four Markets

- Each cell: breach rate of VaR 1% in % / Z_2 p -value / MCS p -value (FZ0)

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	1.67 / <0.001 / <0.001	1.17 / 0.134 / <0.001	0.77 / 0.863 / 0.87	0.51 / 1.000 / <0.001
Normal	2.70 / <0.001 / <0.001	1.90 / <0.001 / <0.001	1.74 / 0.001 / 0.63	1.06 / 0.973 / <0.001
Student-t	2.00 / 0.002 / <0.001	1.58 / 0.003 / <0.001	0.71 / 0.625 / 0.30	0.85 / 0.920 / <0.001
GARCH-t	1.77 / <0.001 / 0.01	1.46 / <0.001 / 0.09	1.12 / 0.050 / 0.84	1.17 / 0.630 / 0.90
FHS	1.23 / 0.172 / 1.00	1.20 / 0.188 / 0.90	0.92 / 0.345 / 0.87	0.85 / 0.968 / 0.99
GARCH-EVT	1.13 / 0.090 / 0.21	0.99 / 0.210 / 1.00	0.95 / 0.581 / 1.00	0.78 / 0.976 / 1.00

- Only FHS and GARCH-EVT pass the Z_2 test at 5% and stay in the 90% MCS in all four markets; on Bitcoin every model stays in the MCS, so the comparison there is inconclusive

Appendix: The Discovery Loop (1/2): Prompts

- Prompts (tools: Semantic Scholar, Elicit, Google Scholar; check every DOI on Crossref)
 - ▶ List peer-reviewed work since 2019 on conformal prediction under serial dependence, with DOI, exact guarantee and assumptions.
 - ▶ Give three rival explanations for crisis over-breaches of split conformal VaR and what each predicts for breach rates by regime.
 - ▶ Act as a hostile referee: list the strongest reasons this crisis effect is an artefact and the test that would settle each.

Appendix: The Discovery Loop (2/2): Checks the Researcher Runs

- **Look-ahead bias**
 - ▶ change the observations after day $t - 1$ and confirm that VaR_t , the scale and the crisis flag do not move
- **Dependent breaches**
 - ▶ Christoffersen and duration tests, or a block bootstrap of the hit sequence
- **Power**
 - ▶ 613 crisis days give 6.1 expected breaches of VaR 1%
 - ▶ a true 2% rate is detected with probability 57%
- **Multiplicity**: $4 \text{ markets} \times 2 \text{ levels} \times 3 \text{ methods} = 24 \text{ tests}$
 - ▶ Holm controls the FWER (family-wise error rate)
 - ▶ Benjamini–Hochberg controls the FDR (false discovery rate)
- **Misstated theorems**
 - ▶ e.g. “ACI guarantees conditional coverage”
- **Off-by-one quantiles**
 - ▶ $k = \lceil (n + 1)(1 - \alpha) \rceil$ is not the default of `np.quantile`

Appendix: Project Seed, Crisis-Conditional Coverage

- Questions
 - ▶ can a distribution-free VaR 1% and ES 2.5% forecast achieve crisis-conditional coverage?
 - ▶ does this coverage hold in all four markets under a pre-registered test?
- Replicate first: split conformal VaR 5% crisis rates (S&P 500 7.29%, EUR/RON 11.43%) and the ACI share of infinite VaR 1% (13.4%)
- Deliverables
 - ▶ A pre-registration (hypotheses, tests, corrections, power) dated before the first run
 - ▶ A candidate method, e.g. [Gibbs, Cherian & Candès \(2025\)](#) with the volatility regime as conditioning class; VaR tests by regime; ES tested with the ES regression backtest ([Bayer & Dimitriadis \(2022\)](#)) or Z_2
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`:
github.com/danpele/MFM/tree/main/project

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